05 - Text + Image Fusion

Goal: combine the text and image models and check if this beats text alone.

1. Turn the text SVM scores into probabilities.
2. Average the two: the text model gets a weight between 0 and 1, the image model gets the rest.
3. Compare text alone, image alone and fusion on the same 16984 validation products.

1. Text probabilities

We use the optimised text model from notebook 04 (all features, binary TF-IDF, LinearSVC with C = 1). A LinearSVC gives scores, not probabilities. Calibration learns on the training set (3-fold CV) how to turn these scores into probabilities.

```
Text probabilities saved: (16984, 28)
Each row sums to 1: True
```

2. Match text and image on the same products

```
Products scored by both: 16984
```

3. Choose the text weight

We try text weights from 0 (image only) to 1 (text only), in steps of 0.1.

Limitation: image probabilities exist only for validation, so the weight is chosen on validation. The fusion score is therefore slightly optimistic.

Text weight	Weighted F1
0.0000	0.4817
0.1000	0.5446
0.2000	0.6434
0.3000	0.7453
0.4000	0.8095
0.5000	0.8427
0.6000	0.8492
0.7000	0.8501
0.8000	0.8476
0.9000	0.8453
1.0000	0.8430

```
Best text weight: 0.7
```

4. Text vs image vs fusion

Model	Weighted F1	Macro F1
Text only	0.8430	0.8326
Image only	0.4817	0.4230
Fusion (w=0.7)	0.8501	0.8401

Code	Category	Text F1	Fusion F1	Gain
2705	New Books	0.711	0.773	0.062
10	Used Books	0.566	0.610	0.044
40	Games	0.753	0.781	0.028
2403	Book Lots	0.768	0.791	0.023
2280	Magazines	0.810	0.826	0.016
2462	Used Games	0.825	0.838	0.013
1160	Cards	0.954	0.965	0.011
1140	Figurines	0.830	0.839	0.009
1940	Food	0.927	0.936	0.009
1281	Board Games	0.616	0.625	0.009
2582	Garden Furniture	0.768	0.775	0.006
50	Game Accessories	0.836	0.841	0.005
2585	Garden Tools	0.833	0.837	0.004
2522	Stationery	0.944	0.946	0.002
1560	Furniture	0.852	0.852	0.000
1301	Baby Socks	0.972	0.972	0.000
2905	PC Games	0.988	0.988	0.000
2220	Pets	0.883	0.883	-0.001
1302	Outdoor Toys	0.837	0.836	-0.001
2060	Decor	0.828	0.827	-0.001
1320	Baby	0.862	0.860	-0.002
2583	Pool	0.981	0.978	-0.003
60	Consoles	0.879	0.876	-0.003
1920	Textiles	0.920	0.917	-0.003
1280	Toys	0.712	0.707	-0.005
1300	RC Models	0.926	0.919	-0.007
1180	Wargames	0.697	0.685	-0.012

Analysis

- Text alone (calibrated): weighted F1 0.8430. Image alone: 0.4817.
- Fusion with 70% text and 30% image: weighted F1 0.8501 (+0.7 points) and macro F1 0.8401 (+0.8 points).
- The weight matters little between 0.6 and 0.8 (0.8476 to 0.8501).
- Book categories gain the most: New Books +6.2 points, Used Books +4.4, Book Lots +2.3, Magazines +1.6. These are the categories where text is weakest.
- No category loses much. The largest loss is Wargames (-1.2 points).
- Limitation: the weight was chosen on validation, so the gain is slightly optimistic. A clean version needs image predictions on train to choose the weight with cross-validation.

Conclusion

Combining text and image gives the best model so far: 0.8501 weighted F1 on validation, against 0.8430 for text alone. Images help mainly on book categories.